Charting "AI" in the Course Description Archive for Research

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The recent rapid changes in AI have been mirrored by equally rapid shifts in the classroom: given the speed at which a wide range of things called "AI" have entered all facets of the classroom we face an urgent need to document and understand the evolving AI education landscape. While researchers across fields and disciplines seem to agree that there is an AI "boom" in higher education, research in this area has been limited by the data sources available. Scoping reviews on the subject turn to the general-purpose search engines and databases available to them (Bond et al. 2024; Maphosa / Maphosa 2023). Other research looks at specific sites of practice or student/instructor interview/survey data (Shaw et al. 2023). Published research on education, however, only accounts for a small sliver of the actual teaching landscape, and the speed at which AI is changing education requires new, and frequently updated, data sources. Our project text-mines course descriptions and catalog information to map how postsecondary educational institutions are confronting "AI" in the classroom. Our use of student-facing course descriptions leverages the rate at which these documents are maintained, updated, and revised as institutions conduct the day-to-day, term-to-term work of providing instruction.

A joint effort of Humanities and Education researchers, our project aims to supplement ongoing research on the intersection of AI and education. For example, the current HAI AI Index Report tracks change at the degree program level but does not capture smaller-scale reactions to AI —missing both the single elective course taken by an interested student and things like the addition of "AI" to an existing course description on tech ethics (Maslej et al. 2024). Our project recognizes that there is a large community of educational practitioners, many of whom are solving the same problems and tackling the same big questions, across the nation, and that there is an even larger community of students who receive information about their educational pathways—the purpose, structure, and demands of those pathways—through the course description.

Course descriptions are a unique academic genre that often reflects the competing priorities of institutions, educators, and societal needs. They are succinct, prosaic pieces of writing that act as the public-facing representations of our teaching work. They serve as decision-making tools for students, guide new instructors, and inform department approvals and legislative reviews. Despite their significance, course descriptions are rarely archived systematically, and some institutions regularly remove them from their websites. This removal creates a gap in available data, making it difficult for researchers to trace curricular trends over time. To begin to address this discontinuity, this project contributes to the development of the Course Description Archive for Research (CDAR), an interdisciplinary project designed to preserve and analyze course descriptions.

CDAR aims to address the growing gap in accessible historical course data in the United States by systematically web scraping and web archiving course descriptions from higher education institutions. By establishing an accessible and citable repository, CDAR aims to create a living resource that provides insight into the evolution of higher education curricula and supports ongoing research in the field. Our current pilot study for CDAR captures the University of California and the California State University systems' course catalogs for all departments, focusing on data for Academic Year 24-25.

In our analysis, we use computational text analysis and qualitative coding to explore how, when, and where "AI" and associated concepts/methods (like "LLM") appear in course descriptions. Due to their generic form, course descriptions let us glim-

pse how AI is rhetorically positioned in ways that other classroom artifacts like syllabi, and broader documents like degree-program learning outcomes, do not. As our course-description data shows, AI can take on any number of roles in a given course offering. Therefore, looking for patterns within the course descriptions themselves will help us understand how American educational institutions are addressing AI in the classroom. Our survey follows linguistic, thematic, and institutional lines of inquiry. What verbs are associated with AI and the students who take these courses? Is AI a tool (we will **use** AI to …), a space for critique (how can we **understand** AI), or a force shaping society (AI will **impact** …)? What if "AI" is paired with other key words? Is "AI" more common in humanities or STEM coursework? What about "large language model"? Does it matter if the school has a DH program, or offers a degree in data science? We pair our text-mining with institutional metadata, including diversity of the student body, the number and kinds of degree programs offered, and physical location to begin to map this rapidly-changing educational landscape, surfacing an interdisciplinarity that does not appear when we look at changes in top-level degree program descriptions.

For example, at University of California, Berkeley, 52 courses this semester mention one of the following terms: AI, artificial intelligence, (large) language model, natural language processing, and NLP. When paired with the keyword "ethic," these words emerge embedded in close proximity to topics ranging from "changing landscapes of labor, industry, and city life" to "privacy" and "questions of morality." When considered with "tools," we can highlight a constellation of verbs that include "using," "summarizing," "augmenting," and "expanding." These early results suggest that questions around the ethics of "AI" are highly variegated, while wide-reaching across society; and that conceptions of "AI" as a tool are underscored by some degree of medium specificity, and particular methods of transformations.

Public and higher education not only train our public in how to engage with the humanities; they also stage the constant renegotiation of humanistic values and methods at national, state, and hyper-local scales. The changes we see in the discourse of "AI" and humanistic ethics in higher and public education are essential for understanding the future of the role of the digital humanities in education and for the public. We see our text-mining questions on AI and the broader research agenda facilitated by the development of CDAR as working towards this understanding.

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